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Black Holes and Quantum Loops: More Than Just a Game

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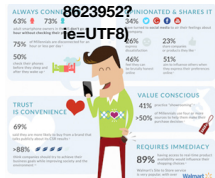
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(http://commons.wikipedia.org/wiki/File:250GB_Slim_PS3.jpg)



Khanna was not by any means a Sony Playstation 3 aficionado. But like all scientists who depend on computers, he keeps up on CPU technology, and when he heard that Sony's industry-leading game station employed a new Cell Broadband engine, he quickly realized how economically he could put together his own supercomputer.

Khanna, who was born and grew up in Northern India, earned his Ph.D at Penn State (<http://www.psu.edu/>) and is now an associate professor of physics at University of Massachusetts, Dartmouth (<http://gravity.phy.umassd.edu/main.html>). His chief areas of interest are black holes and quantum loop gravity, a new theory of gravitation that reworks Einstein's general theory of relativity in terms of quantum loops.

"The Playstation 3 offers two significant advantages in this regard," he told me. "One: Its Cell processor is extremely powerful... and Two: the PS3 hardware is deeply discounted. The combination of these two facts offers one the option of doing ten times more science for each dollar spent as compared with traditional computer systems."

For anyone in research and development, it costs money to rent time on a supercomputer, and it costs time as well—waiting for your turn after the other researchers who applied for access ahead of you. Because of the PS3s Cell processor and its consumer-level price, for the rental cost associated with a single simulation, Khanna was able to network eight Playstation 3s to run the same calculation. And there was the added benefit of being able to run the calculation again at no extra cost.

When he finally got Sony to realize what he wanted to do with their game-stations, the R&D people got excited. They provided him with four of his initial eight machines, and technical support. After the success of his initial simulations, the university dean helped him double the size of his set-up. The National Science Foundation provided three grants to support his simulations as well.

“In essence, my 16 PS3 system performs at the rate of 100+ processor cores of a traditional workstation processor,” said Khanna. “That yields a nearly ten-fold cost effectiveness (from the viewpoint of performance-per-dollar spent and also the performance-per-Watt consumed by the system).”

Other research groups have taken up the Playstation approach to simulation, at [University of New Hampshire](http://www.eos.unh.edu/news/0708/cluster.shtml) (<http://www.eos.unh.edu/news/0708/cluster.shtml>), the [Rowland Institute](http://www.rowland.harvard.edu/rjf/cox/Projects/projects/Projects/page10.html) (<http://www.rowland.harvard.edu/rjf/cox/Projects/projects/Projects/page10.html>) at Harvard, and even the U.S. Air Force [Research Laboratory](http://www.stripes.com/military-life/turning-playstations-into-power-stations-1.99009) (<http://www.stripes.com/military-life/turning-playstations-into-power-stations-1.99009>) in New York.

The latest results of Khanna’s self-built computer network have born fruit, for just one example, in a [recent paper](http://xxx.lanl.gov/abs/1003.0485) (<http://xxx.lanl.gov/abs/1003.0485>) he co-authored with two colleagues at MIT on how two black holes could merge into one. Einstein’s field equations for general relativity are notoriously labor intensive, even for just modeling a single star, let alone a collapsed star or a pair of them rotating slowly into each other. Writing the code to run the calculations presents challenges even before getting to the stage of actually running them in a computer simulation.

“The PS3’s processor — Cell BE — is a difficult beast to program,” Khanna said. “Its radically new, heterogeneous design, is very different from other processors to date.” That said, in the time since he first adopted the

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playstations for his use, several software development kits have emerged from Sony as well as companies like IBM and other third parties that have made the programming model a little simpler. “Yes, things have gotten a little easier in that context,” he said.

In addition to inspiring his recent papers, Khanna expects that his black hole simulation work will provide real support to two current experiments: the Laser Interferometer Gravitational Wave Observatory ([LIGO](http://www.ligo.caltech.edu/) (<http://www.ligo.caltech.edu/>)), which is managed by Cal Tech and MIT, and NASA’s planned launch of the Laser Interferometer Space Antenna ([LISA](http://lisa.jpl.nasa.gov/) (<http://lisa.jpl.nasa.gov/>)). Both of these large scale efforts are designed to detect gravitational waves, the signature ‘radiation’ of gravity, predicted by general relativity.

“The two projects focus on different sources of gravitational waves,” according to Khanna. “LIGO looks for bursts of gravitational waves coming from the merger of (roughly) two- solar-mass black holes, whereas LISA will be looking for these waves coming from the so-called extreme-mass-ratio-inspiral (EMRI) case, i.e. the capture of a relatively small solar-mass star or black hole by a supermassive black hole.

“Supermassive black holes lurk at the center of most galaxies and routinely devour smaller objects like stars and other black holes. These processes will be the main focus of the European Space Agency and NASA LISA mission. These are very interesting processes because the gravitational waves from such systems encode a lot of information about the central large black hole itself and will perhaps give us extremely detailed information about the nature of black holes in general.”

Khanna and his Playstation network are broadening their application. This year he received a grant from the [Foundational Questions Institute](http://www.fqxi.org/grants/large/awardees/view/details/2008/khanna) (FQXi) in New York, to study models of Quantum Gravity, a subject that on its face seems quite distinct from the black holes and gravitational waves. But Khanna realized that some of the computational methodology is similar between both projects, so it made sense for him to apply his Playstation cluster technique to the quantum gravity.

“The quantum gravity project has to do with exploring the nature of the universe very close to the moment of the Big Bang,” he said. “Perhaps even exploring the ‘past’ of the moment of ‘beginning’ itself. That is very exciting research on its own, but one that is perhaps more speculative than the very concrete black hole astrophysics research I do. FQXi is very interested in supporting such fundamental research topics and they were kind enough to provide me some modest research support to advance that quantum gravity project.”

One of the issues many physicists hope that a deeper study of quantum gravity

will resolve, is whether the Big Bang really is the ultimate origin of the universe. Or was there perhaps a universe that existed before ours, one that expanded, developed, and then collapsed before ‘bouncing’ out again into the cosmos we live in? No small number of recent popular science books have treated the subject.

But they are all speculative, Khanna points out. “The issue is that there are too many alternative models! That is the serious challenge of working in an area like quantum gravity that currently lacks confirmation through observation, or experimentation. Most ideas are simply speculation. And there are a lot of very interesting but speculative models out there!”

Perhaps Khanna’s Playstation simulations of quantum gravity will provide support some day for a future space-based experiment to look for signs beyond the beginning.



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